

INFO8006 Introduction to Artificial Intelligence

Exercise session 3: Quantifying uncertainty

RECAP

Definitions

A probability is a degree of belief in a proposition.

Sample space	Ω , of possible worlds ω
Axioms	$0 \leq P(\omega) \leq 1$, $\sum_{\omega \in \Omega} P(\omega) = 1$
Event	$A \subseteq \Omega$, $P(A) = \sum_{\omega \in A} P(\omega)$
Random variable	$X : \Omega \rightarrow D_X$, with $X = x$ the event $\{\omega \in \Omega : X(\omega) = x\}$
Distribution	$\mathbf{P}(X)$, the numbers $P(x) = P(X = x)$ for $x \in D_X$
Joint distribution	$\mathbf{P}(X_1, \dots, X_n)$, the numbers $P(x_1, \dots, x_n)$
Conditional probability	$P(a b) = P(a, b) / P(b)$, for $P(b) > 0$
Conditional distribution	$\mathbf{P}(X Y)$, one $\mathbf{P}(X y)$ for each $y \in D_Y$

Random variables are written in upper case and their values in lower case; a boolean variable A takes the values a and $\neg a$. $P(x)$ is a number, $\mathbf{P}(X)$ a table.

Rules

Complement	$P(\neg a) = 1 - P(a)$
Union	$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$
Product rule	$P(a, b) = P(a b)P(b) = P(b a)P(a)$
Sum rule, total probability	$P(x) = \sum_y P(x, y) = \sum_y P(x y)P(y)$
Chain rule	$P(x_{1:n}) = \prod_i P(x_i x_{1:i-1})$
Bayes' rule	$P(a b) = P(b a)P(a) / P(b)$
Independence, $X \perp Y$	$P(x, y) = P(x)P(y)$
Conditional independence, $X \perp Y Z$	$P(x, y z) = P(x z)P(y z)$
Naive Bayes	$P(c, e_{1:n}) = P(c) \prod_i P(e_i c)$

The rules hold for all values, and between distributions. Naive Bayes assumes the effects $e_{1:n}$ conditionally independent given the cause c . The joint distribution answers every query: select the entries consistent with the evidence, sum out the other variables, normalize.

Bayes' rule

A model is given in the *causal* direction: for an agent, a *prior* $\mathbf{P}(S)$ over the unobserved state and a *sensor model* (or *likelihood*) $\mathbf{P}(e | S)$ of its percepts. The *diagnostic* question is answered by the *belief*, or *posterior*, $\mathbf{P}(S | e) = \frac{1}{Z} \mathbf{P}(e | S) \mathbf{P}(S)$, where $Z = P(e) = \sum_s P(e | s)P(s)$ is the probability of the *evidence*. For percepts conditionally independent given the state, each belief is the prior of the next, $\mathbf{P}(S | e_{1:k+1}) = \frac{1}{Z} \mathbf{P}(e_{k+1} | S) \mathbf{P}(S | e_{1:k})$.

In session: exercises 1, 2, 3, 5.

Exercise 1 Rational beliefs (AIMA ch. 12)

Let $A, B \subseteq \Omega$ be two *events*. An agent holds the beliefs $P(A) = 0.4$ and $P(B) = 0.3$. What ranges of probabilities would it be rational for the agent to hold for the events $A \cup B$ and $A \cap B$? What if the agent also believes that $P(A | B) = 0.5$?

Solution.

1. Since $P(E) = \sum_{\omega \in E} P(\omega)$ sums nonnegative terms, $E \subseteq F$ implies $P(E) \leq P(F)$. With $A, B \subseteq A \cup B$ and $A \cap B \subseteq A, B$, and the union rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, the agent's beliefs should satisfy

$$\begin{aligned} 0.4 &= \max\{P(A), P(B)\} \leq P(A \cup B) \leq \min\{1, P(A) + P(B)\} = 0.7 \\ 0 &= \max\{0, P(A) + P(B) - 1\} \leq P(A \cap B) \leq \min\{P(A), P(B)\} = 0.3. \end{aligned}$$

The lower bound on $P(A \cup B)$ and the upper bound on $P(A \cap B)$ follow from monotonicity; the other two from the union rule, with $P(A \cap B) \geq 0$ and $P(A \cup B) \leq 1$. The two beliefs are not independent choices: by the union rule, $P(A \cup B) + P(A \cap B) = 0.7$. The extremes are $B \subseteq A$, with $P(A \cup B) = 0.4$ and $P(A \cap B) = 0.3$, and disjoint events, with 0.7 and 0.

2. If the agent believes that $P(A | B) = 0.5$, it should also believe that

$$P(A \cap B) = P(A | B)P(B) = 0.5 \times 0.3 = 0.15,$$

by the product rule. Then, the agent should rationally believe that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.3 - 0.15 = 0.55.$$

Exercise 2 Full joint distribution (AIMA ch. 12)

A monitoring program supervises the bearings of a wind turbine. A worn bearing (*fault*) tends to make the turbine vibrate (*vibration*) and to trigger the alarm of a diagnostic probe (*alarm*). The program holds the joint distribution $\mathbf{P}(V, A, F)$ below, over the boolean random variables V (vibration), A (alarm) and F (fault). Compute the following probabilities.

	v		$\neg v$	
	a	$\neg a$	a	$\neg a$
f	0.108	0.012	0.072	0.008
$\neg f$	0.016	0.064	0.144	0.576

- (a) $P(v)$

Solution.

$$P(v) = \sum_{a'} \sum_{f'} P(v, a', f') = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

- (b) $P(f)$

Solution.

$$P(f) = \sum_{v'} \sum_{a'} P(v', a', f) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$$

(c) $P(v \mid f)$

Solution.

$$P(v \mid f) = \frac{P(v, f)}{P(f)} = \frac{\sum_{a'} P(v, a', f)}{P(f)} = \frac{0.108 + 0.012}{0.2} = 0.6$$

(d) $P(f \mid v \vee a)$

Solution.

$$\begin{aligned} P(f \mid v \vee a) &= \frac{P(f, v \vee a)}{P(v \vee a)} = \frac{P(f) - P(\neg v, \neg a, f)}{1 - P(\neg v, \neg a)} \\ &= \frac{0.2 - 0.008}{1 - 0.584} = 0.4615 \end{aligned}$$

(e) Are V and A independent? Are they conditionally independent given F ?

Solution. With $P(a) = 0.108 + 0.016 + 0.072 + 0.144 = 0.34$, $P(v, a) = 0.108 + 0.016 = 0.124 \neq P(v)P(a) = 0.2 \times 0.34 = 0.068$: V and A are not independent. Given F , for boolean variables it suffices to check one entry per value of F , the others following by complement:

$$\begin{aligned} P(v, a \mid f) &= \frac{0.108}{0.2} = 0.54 = P(v \mid f)P(a \mid f) = 0.6 \times 0.9, \\ P(v, a \mid \neg f) &= \frac{0.016}{0.8} = 0.02 = P(v \mid \neg f)P(a \mid \neg f) = 0.1 \times 0.2, \end{aligned}$$

so $V \perp A \mid F$. The table is a naive Bayes model, with the fault as the cause and the vibration and the alarm as its effects: $\mathbf{P}(V, A, F) = \mathbf{P}(F)\mathbf{P}(V \mid F)\mathbf{P}(A \mid F)$ needs $1 + 2 + 2 = 5$ numbers instead of 7. Vibration and alarm are dependent only through their common cause.

Exercise 3 Fraud detection (AIMA ch. 12)

A fraud detection program screens card payments. It is 99% accurate, *i.e.* it flags a fraudulent payment with probability 0.99 and a legitimate payment with probability 0.01. Fraud is rare: it strikes only 1 payment in 10 000. The program has just flagged your payment. Why is it good news, for you, that fraud is rare? What is the probability that the payment is actually fraudulent?

Solution. Let F and A be the boolean random variables telling whether the payment is fraudulent and whether it is flagged. For the program, F is the unobserved state and A the percept, with the prior $P(f) = 10^{-4}$ and the sensor model $P(a \mid f) = 0.99$, $P(a \mid \neg f) = 0.01$. Its belief follows from Bayes' rule,

$$\begin{aligned} P(f \mid a) &= \frac{P(a \mid f)P(f)}{P(a)} \\ &= \frac{P(a \mid f)P(f)}{P(a \mid f)P(f) + P(a \mid \neg f)P(\neg f)} \\ &= \frac{0.99 \times 10^{-4}}{0.99 \times 10^{-4} + 0.01 \times (1 - 10^{-4})} = 0.0098. \end{aligned}$$

Rarity is good news because the prior $P(f)$ is small. Out of 1 000 000 payments, 100 are fraudulent and 99 of them are flagged, while 999 900 are legitimate and 9999 of them are flagged by mistake. Only 99 of the 10 098 flagged payments, about 1%, are fraudulent.

Exercise 4 Conditional independence (AIMA ch. 12)

For each of the following statements, about values a, b, c of the random variables A, B, C , either prove it is valid or give a counterexample. Conditional probabilities are only taken given events of positive probability, as their definition requires.

- (a) If $P(a, b, c) > 0$ and $P(a | b, c) = P(b | a, c)$, then $P(a | c) = P(b | c)$.

Solution. True. By the product rule, $P(a | b, c)P(b, c) = P(a, b, c) = P(b | a, c)P(a, c)$. Since $P(a, b, c) > 0$, $P(a | b, c) = P(b | a, c) > 0$, and dividing by it gives $P(b, c) = P(a, c)$ and thereby $P(b | c) = P(a | c)$.

- (b) If $P(a | b, c) = P(a)$, then $P(b | c) = P(b)$.

Solution. False. The statement $P(a | b, c) = P(a)$ says that A is independent of B and C jointly; it makes no claim about the dependence between B and C . Counterexample: A and B are two independent fair coin flips, and $C = B$. Taking the value true for a, b and c , $P(a | b, c) = P(a) = 1/2$, but $P(b | c) = 1 \neq P(b) = 1/2$.

- (c) If $P(a | b) = P(a)$, then $P(a | b, c) = P(a | c)$.

Solution. False. The statement $P(a | b) = P(a)$ says that A is independent of B , which does not imply that A is conditionally independent of B given C . Counterexample: A and B are two independent fair coin flips, and $C = A \vee B$. Taking the value true for a, b and c , $P(a | b) = P(a) = 1/2$, but $P(a | b, c) = P(a, b)/P(b) = 1/2 \neq P(a | c) = P(a)/P(c) = (1/2)/(3/4) = 2/3$.

Exercise 5 Driving styles (AIMA ch. 13)

A self-driving car shares the highway with another car, whose driving style it does not know. It models three styles, cautious (a), normal (b) and aggressive (c), equally likely a priori, under which the other car changes lanes during a given minute with probabilities 0.2, 0.6 and 0.8, respectively. The self-driving car observes the other car for three minutes, recording whether it changes lanes (l) or keeps its lane (k) during each minute, as X_1, X_2 and X_3 . Let Y be the style of the other car.

- (a) Write the joint distribution $\mathbf{P}(Y, X_1, X_2, X_3)$ of this model as a product of smaller tables, and give these tables.

Solution. The process is described with the following random variables:

- $Y \in \{a, b, c\}$, the driving style of the other car;
- $X_i \in \{l, k\}$, its behaviour during the i -th minute.

The observations are conditionally independent given the style, so the model is a naive Bayes model, with the style as the cause and the observations as its effects:

$$\mathbf{P}(Y, X_1, X_2, X_3) = \mathbf{P}(Y)\mathbf{P}(X_1 | Y)\mathbf{P}(X_2 | Y)\mathbf{P}(X_3 | Y),$$

with the same table $\mathbf{P}(X_i | Y)$ for the three minutes:

y	$P(y)$	$P(X_i = l y)$
a	1/3	0.2
b	1/3	0.6
c	1/3	0.8

The two tables hold $2 + 3 = 5$ free numbers, against $3 \times 2^3 - 1 = 23$ for the full joint distribution.

- (b) Determine the most likely driving style if the other car changed lanes during the first two minutes and kept its lane during the third.

Solution. The style Y is the unobserved state and the X_i are the percepts, with the prior $\mathbf{P}(Y)$ and the sensor model $\mathbf{P}(X_i | Y)$ of (a). The belief of the car is the posterior $\mathbf{P}(Y | X_1 = l, X_2 = l, X_3 = k)$. By Bayes' rule,

$$\mathbf{P}(Y | X_1 = l, X_2 = l, X_3 = k) = \frac{1}{Z} \mathbf{P}(X_1 = l, X_2 = l, X_3 = k | Y) \mathbf{P}(Y).$$

With the factorization of (a),

$$P(y | X_1 = l, X_2 = l, X_3 = k) = \frac{1}{Z} \times \frac{1}{3} \times \begin{cases} 0.2 \times 0.2 \times 0.8 = 0.032 & \text{if } y = a \\ 0.6 \times 0.6 \times 0.4 = 0.144 & \text{if } y = b \\ 0.8 \times 0.8 \times 0.2 = 0.128 & \text{if } y = c, \end{cases}$$

with $Z = \frac{1}{3}(0.032 + 0.144 + 0.128)$. The posterior is $\langle 0.105, 0.474, 0.421 \rangle$: the normal style is the most likely, though the aggressive one is nearly as probable. Finding the most likely style only needs the unnormalized values. The order of the observations does not matter: any two lane changes and one lane keep have the same probability given each style.

- (c) The car updates its belief after each minute, the belief after one minute being the prior of the next. Compute its belief after one, two and three minutes, and compare with (b).

Solution. Starting from the prior $\langle 1/3, 1/3, 1/3 \rangle$ over $\langle a, b, c \rangle$, each minute multiplies the belief by the sensor model and normalizes:

$$\begin{aligned} \mathbf{P}(Y | l) &= \frac{1}{Z} \langle 0.2, 0.6, 0.8 \rangle \times \langle 1/3, 1/3, 1/3 \rangle = \langle 0.125, 0.375, 0.5 \rangle, \\ \mathbf{P}(Y | l, l) &= \frac{1}{Z} \langle 0.2, 0.6, 0.8 \rangle \times \langle 0.125, 0.375, 0.5 \rangle = \langle 0.038, 0.346, 0.615 \rangle, \\ \mathbf{P}(Y | l, l, k) &= \frac{1}{Z} \langle 0.8, 0.4, 0.2 \rangle \times \langle 0.038, 0.346, 0.615 \rangle = \langle 0.105, 0.474, 0.421 \rangle, \end{aligned}$$

where $\times$ multiplies entry by entry. The last belief is the posterior of (b): since the percepts are conditionally independent given the style, updating minute by minute gives the same result as conditioning on the three percepts at once. After two lane changes, the aggressive style is the most likely; keeping the lane during the third minute makes the normal style overtake it.

QUIZ

1. Bayes' rule states that $P(y | x)$ is equal to ...

- ☐ $\frac{P(x | y)P(x)}{P(y)}.$
☒ $\frac{P(x | y)P(y)}{P(x)}.$
☐ $\frac{P(x, y)}{P(y)}.$
☐ $P(x | y)P(y).$

2. Which of the following is true?

- ☐ $P(x, y | z) = \frac{P(x | y, z)}{P(y, z | x)}.$
☐ $P(x, y, z) = P(x)P(y)P(z).$
☒ $P(x, y | z) = P(y | x, z)P(x | z).$
☐ $P(x | y, z) = \frac{P(x, y, z)}{P(y)P(z)}.$

3. The naive Bayes model assumes ...
- ☒ conditional independence between the effects, given the cause.
 - ☐ independence between the effects.
 - ☐ independence between the effects and the cause.
 - ☐ conditional independence between each effect and the cause, given the other effects.
4. With a uniform prior $\mathbf{P}(A)$, the posterior $\mathbf{P}(A \mid b)$ is proportional to ...
- ☐ the prior $\mathbf{P}(A)$.
 - ☒ the likelihood $\mathbf{P}(b \mid A)$.
 - ☐ the evidence $P(b)$.
 - ☐ the joint distribution $\mathbf{P}(A, B)$.